

1D CNN — Architecture Variant Comparison

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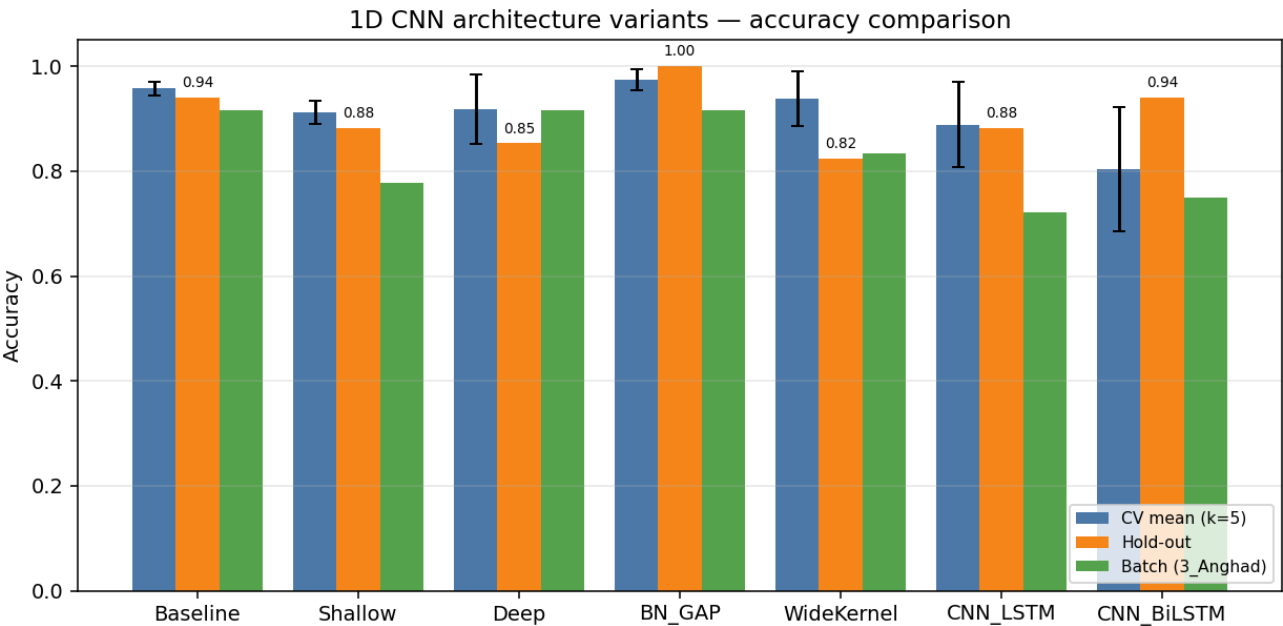
Data & configuration

Training path 1	/home/jestin/ThesisRepo/ML/NewTestData/1_Alan/Dynamic
Training path 2	/home/jestin/ThesisRepo/ML/NewTestData/2_Alex/Dynamic
Training path 3	/home/jestin/ThesisRepo/ML/NewTestData/4_Daniel/Dynamic
Training path 4	/home/jestin/ThesisRepo/ML/NewTestData/6_Harry/Dynamic
Training path 5	/home/jestin/ThesisRepo/ML/NewTestData/8_Jestin/Dynamic
Training path 6	/home/jestin/ThesisRepo/ML/NewTestData/11_Mansh/Dynamic
Training path 7	/home/jestin/ThesisRepo/ML/NewTestData/12_Marcus/Dynamic
Training path 8	/home/jestin/ThesisRepo/ML/NewTestData/14_StephenV2/Dynamic
Training path 9	/home/jestin/ThesisRepo/ML/NewTestData/15_Tash/Dynamic
Training path 10	/home/jestin/ThesisRepo/ML/NewTestData/17_Raquel/Dynamic
Batch-test root	/home/jestin/ThesisRepo/ML/NewTestData/3_Anghad/Dynamic
Total trials	340 (train pool 306, hold-out 34)
Classes	7: Double_Lower, Double_Nothing, Double_Pistol_Recoil, Double_Raise, Double_Wiggle, Double_cmere, Drum_Roll
Sensor channels	176 (sequence length 90)
Resample / filter	90 steps · Butterworth cutoff 15.0 Hz, order 4
Normalisation	minmax
CV folds · shuffle	5 · True
Augmentation	on · copies=5 · amplitude_scale, baseline_offset
Epochs · batch size	15 · 16

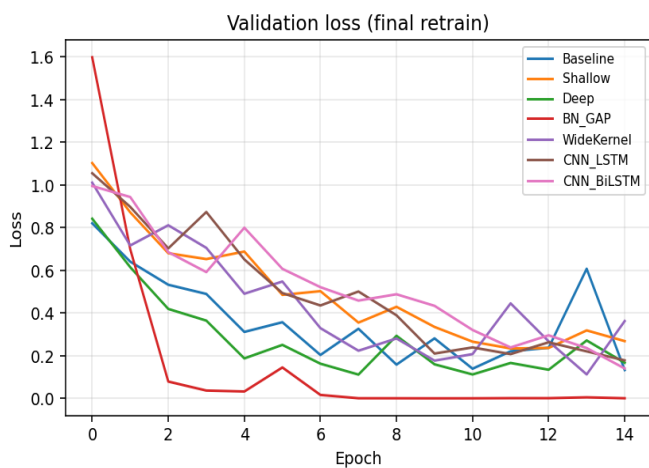
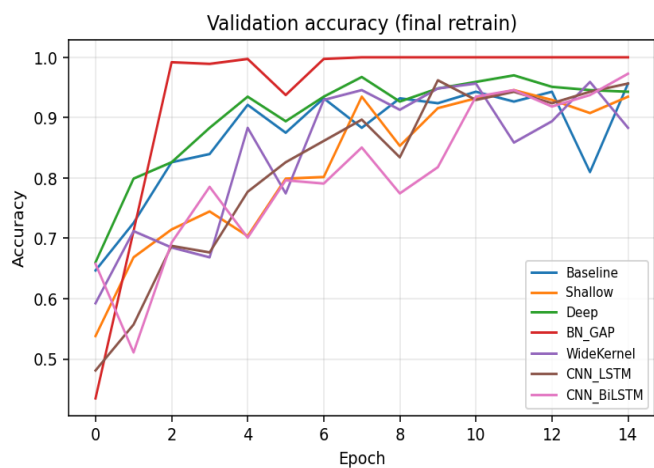
Variant comparison (summary)

Variant	Params	CV mean (k=5)	CV std	Hold-out acc	Hold-out loss	Batch acc	Batch n
Baseline	113,255	0.9575	0.0132	0.9412	0.2125	0.9167	33/36
Shallow	72,487	0.9117	0.0223	0.8824	0.3899	0.7778	28/36
Deep	204,007	0.9181	0.0665	0.8529	0.4182	0.9167	33/36
BN_GAP	57,319	0.9739	0.0195	1.0000	0.0103	0.9167	33/36
WideKernel	131,687	0.9382	0.0523	0.8235	0.5305	0.8333	30/36
CNN_LSTM	68,455	0.8887	0.0814	0.8824	0.2962	0.7222	26/36
CNN_BiLSTM	105,575	0.8039	0.1183	0.9412	0.1788	0.7500	27/36

Best on hold-out: **BN_GAP** · Best on CV: **BN_GAP**



Training curves (final retrain on full training pool)



Variant: Baseline

Conv1D(32,k=4)→Pool→Conv1D(64,k=4)→Pool→Flatten→Dense(64)→Dropout(0.3)→Softmax

Trainable params	113,255
CV mean ± std	0.9575 ± 0.0132
Hold-out test acc / loss	0.9412 · 0.2125
Batch-test acc	0.9167 (33/36)

Cross-validation folds

Fold	Train (post-aug)	Val	Val acc	Best val	Final val
1	1464	62	0.9677	0.9677	0.9677
2	1470	61	0.9672	0.9672	0.9672
3	1470	61	0.9508	0.9508	0.9508
4	1470	61	0.9672	0.9672	0.9672
5	1470	61	0.9344	0.9672	0.9344

Per-class hold-out metrics

Class	Precision	Recall	F1	Support
Double_Lower	1.000	1.000	1.000	5
Double_Nothing	1.000	0.600	0.750	5
Double_Pistol_Recoil	0.833	1.000	0.909	5
Double_Raise	1.000	1.000	1.000	5
Double_Wiggle	0.833	1.000	0.909	5
Double_cmere	1.000	1.000	1.000	5
Drum_Roll	1.000	1.000	1.000	4
macro avg	0.952	0.943	0.938	34
weighted avg	0.951	0.941	0.936	34

Batch test per class (3_Anghad)

Class	Correct	Total	Accuracy
Double_Lower	5	5	1.000
Double_Nothing	2	5	0.400
Double_Pistol_Recoil	5	5	1.000
Double_Raise	5	5	1.000
Double_Wiggle	5	5	1.000
Double_cmere	6	6	1.000
Drum_Roll	5	5	1.000
Overall	33	36	0.917

Variant: Shallow

Single Conv block: Conv1D(32,k=5)→Pool→Flatten→Dense(32)→Dropout(0.3)→Softmax

Trainable params	72,487
CV mean ± std	0.9117 ± 0.0223
Hold-out test acc / loss	0.8824 · 0.3899
Batch-test acc	0.7778 (28/36)

Cross-validation folds

Fold	Train (post-aug)	Val	Val acc	Best val	Final val
1	1464	62	0.9194	0.9355	0.9194
2	1470	61	0.9180	0.9508	0.9180
3	1470	61	0.9344	0.9344	0.9344
4	1470	61	0.9180	0.9344	0.9180
5	1470	61	0.8689	0.9180	0.8689

Per-class hold-out metrics

Class	Precision	Recall	F1	Support
Double_Lower	1.000	1.000	1.000	5
Double_Nothing	0.750	0.600	0.667	5
Double_Pistol_Recoil	0.833	1.000	0.909	5
Double_Raise	0.833	1.000	0.909	5
Double_Wiggle	0.833	1.000	0.909	5
Double_cmere	1.000	0.800	0.889	5
Drum_Roll	1.000	0.750	0.857	4
macro avg	0.893	0.879	0.877	34
weighted avg	0.890	0.882	0.878	34

Batch test per class (3_Angbad)

Class	Correct	Total	Accuracy
Double_Lower	4	5	0.800
Double_Nothing	2	5	0.400
Double_Pistol_Recoil	3	5	0.600
Double_Raise	5	5	1.000
Double_Wiggle	4	5	0.800
Double_cmere	6	6	1.000
Drum_Roll	4	5	0.800
Overall	28	36	0.778

Variant: Deep

Three Conv blocks: Conv1D(32,4)→Conv1D(64,4)→Conv1D(128,3)→Flatten→Dense(128)→Dropout(0.4)→Softmax

Trainable params	204,007
CV mean ± std	0.9181 ± 0.0665
Hold-out test acc / loss	0.8529 · 0.4182
Batch-test acc	0.9167 (33/36)

Cross-validation folds

Fold	Train (post-aug)	Val	Val acc	Best val	Final val
1	1464	62	0.9677	0.9677	0.9677
2	1470	61	0.9836	1.0000	0.9836
3	1470	61	0.8033	0.9672	0.8033
4	1470	61	0.9508	0.9836	0.9508
5	1470	61	0.8852	0.9508	0.8852

Per-class hold-out metrics

Class	Precision	Recall	F1	Support
Double_Lower	0.833	1.000	0.909	5
Double_Nothing	0.500	0.200	0.286	5
Double_Pistol_Recoil	1.000	1.000	1.000	5
Double_Raise	1.000	1.000	1.000	5
Double_Wiggle	0.625	1.000	0.769	5
Double_cmere	1.000	1.000	1.000	5
Drum_Roll	1.000	0.750	0.857	4
macro avg	0.851	0.850	0.832	34
weighted avg	0.847	0.853	0.831	34

Batch test per class (3_Anghad)

Class	Correct	Total	Accuracy
Double_Lower	5	5	1.000
Double_Nothing	2	5	0.400
Double_Pistol_Recoil	5	5	1.000
Double_Raise	5	5	1.000
Double_Wiggle	5	5	1.000
Double_cmere	6	6	1.000
Drum_Roll	5	5	1.000
Overall	33	36	0.917

Variant: BN_GAP

Conv+BN+ReLU ×3 with GlobalAveragePooling1D head — fewer params, less overfit

Trainable params	57,319
CV mean ± std	0.9739 ± 0.0195
Hold-out test acc / loss	1.0000 · 0.0103
Batch-test acc	0.9167 (33/36)

Cross-validation folds

Fold	Train (post-aug)	Val	Val acc	Best val	Final val
1	1464	62	0.9516	0.9516	0.9516
2	1470	61	1.0000	1.0000	1.0000
3	1470	61	0.9508	0.9672	0.9508
4	1470	61	0.9836	0.9836	0.9836
5	1470	61	0.9836	0.9836	0.9836

Per-class hold-out metrics

Class	Precision	Recall	F1	Support
Double_Lower	1.000	1.000	1.000	5
Double_Nothing	1.000	1.000	1.000	5
Double_Pistol_Recoil	1.000	1.000	1.000	5
Double_Raise	1.000	1.000	1.000	5
Double_Wiggle	1.000	1.000	1.000	5
Double_cmere	1.000	1.000	1.000	5
Drum_Roll	1.000	1.000	1.000	4
macro avg	1.000	1.000	1.000	34
weighted avg	1.000	1.000	1.000	34

Batch test per class (3_Anghad)

Class	Correct	Total	Accuracy
Double_Lower	5	5	1.000
Double_Nothing	2	5	0.400
Double_Pistol_Recoil	5	5	1.000
Double_Raise	5	5	1.000
Double_Wiggle	5	5	1.000
Double_cmere	6	6	1.000
Drum_Roll	5	5	1.000
Overall	33	36	0.917

Variant: WideKernel

Conv1D(32,k=8)→Pool→Conv1D(64,k=8)→Pool→Flatten→Dense(64)→Dropout(0.3)→Softmax

Trainable params	131,687
CV mean ± std	0.9382 ± 0.0523
Hold-out test acc / loss	0.8235 · 0.5305
Batch-test acc	0.8333 (30/36)

Cross-validation folds

Fold	Train (post-aug)	Val	Val acc	Best val	Final val
1	1464	62	0.8387	0.9516	0.8387
2	1470	61	0.9836	0.9836	0.9836
3	1470	61	0.9672	0.9672	0.9672
4	1470	61	0.9672	0.9672	0.9672
5	1470	61	0.9344	0.9508	0.9344

Per-class hold-out metrics

Class	Precision	Recall	F1	Support
Double_Lower	0.714	1.000	0.833	5
Double_Nothing	1.000	0.200	0.333	5
Double_Pistol_Recoil	1.000	0.800	0.889	5
Double_Raise	0.833	1.000	0.909	5
Double_Wiggle	0.714	1.000	0.833	5
Double_cmere	1.000	0.800	0.889	5
Drum_Roll	0.800	1.000	0.889	4
macro avg	0.866	0.829	0.797	34
weighted avg	0.868	0.824	0.794	34

Batch test per class (3_Angbad)

Class	Correct	Total	Accuracy
Double_Lower	5	5	1.000
Double_Nothing	1	5	0.200
Double_Pistol_Recoil	3	5	0.600
Double_Raise	5	5	1.000
Double_Wiggle	5	5	1.000
Double_cmere	6	6	1.000
Drum_Roll	5	5	1.000
Overall	30	36	0.833

Variant: CNN_LSTM

Conv1D(32,4)→Pool→Conv1D(64,4)→Pool→Dropout→LSTM(64)→Dense(64)→Dropout→Softmax

Trainable params	68,455
CV mean ± std	0.8887 ± 0.0814
Hold-out test acc / loss	0.8824 · 0.2962
Batch-test acc	0.7222 (26/36)

Cross-validation folds

Fold	Train (post-aug)	Val	Val acc	Best val	Final val
1	1464	62	0.9516	0.9516	0.9516
2	1470	61	0.9344	0.9508	0.9344
3	1470	61	0.7377	0.8197	0.7377
4	1470	61	0.8689	0.8689	0.8689
5	1470	61	0.9508	0.9672	0.9508

Per-class hold-out metrics

Class	Precision	Recall	F1	Support
Double_Lower	1.000	1.000	1.000	5
Double_Nothing	1.000	0.600	0.750	5
Double_Pistol_Recoil	0.833	1.000	0.909	5
Double_Raise	0.714	1.000	0.833	5
Double_Wiggle	1.000	1.000	1.000	5
Double_cmere	1.000	0.600	0.750	5
Drum_Roll	0.800	1.000	0.889	4
macro avg	0.907	0.886	0.876	34
weighted avg	0.910	0.882	0.876	34

Batch test per class (3_Angbad)

Class	Correct	Total	Accuracy
Double_Lower	5	5	1.000
Double_Nothing	1	5	0.200
Double_Pistol_Recoil	4	5	0.800
Double_Raise	5	5	1.000
Double_Wiggle	0	5	0.000
Double_cmere	6	6	1.000
Drum_Roll	5	5	1.000
Overall	26	36	0.722

Variant: CNN_BiLSTM

Conv1D(32,4)→Pool→Conv1D(64,4)→Pool→Dropout→BiLSTM(64)→Dense(64)→Dropout→Softmax

Trainable params	105,575
CV mean ± std	0.8039 ± 0.1183
Hold-out test acc / loss	0.9412 · 0.1788
Batch-test acc	0.7500 (27/36)

Cross-validation folds

Fold	Train (post-aug)	Val	Val acc	Best val	Final val
1	1464	62	0.8226	0.9194	0.8226
2	1470	61	0.8033	0.8852	0.8033
3	1470	61	0.5902	0.8852	0.5902
4	1470	61	0.9508	0.9672	0.9508
5	1470	61	0.8525	0.9016	0.8525

Per-class hold-out metrics

Class	Precision	Recall	F1	Support
Double_Lower	1.000	1.000	1.000	5
Double_Nothing	0.800	0.800	0.800	5
Double_Pistol_Recoil	1.000	0.800	0.889	5
Double_Raise	1.000	1.000	1.000	5
Double_Wiggle	0.833	1.000	0.909	5
Double_cmere	1.000	1.000	1.000	5
Drum_Roll	1.000	1.000	1.000	4
macro avg	0.948	0.943	0.943	34
weighted avg	0.946	0.941	0.941	34

Batch test per class (3_Anghad)

Class	Correct	Total	Accuracy
Double_Lower	5	5	1.000
Double_Nothing	0	5	0.000
Double_Pistol_Recoil	2	5	0.400
Double_Raise	5	5	1.000
Double_Wiggle	4	5	0.800
Double_cmere	6	6	1.000
Drum_Roll	5	5	1.000
Overall	27	36	0.750